

CRUD Operations Using MongoDB

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1. Program:

```
package crud_operations;

import com.mongodb.client.*;
import com.mongodb.client.model.Filters;
import com.mongodb.client.model.Updates;
import org.bson.Document;

public class Crud_inMongoDB {

    public static void main(String[] args) {

        // Connect to MongoDB

        MongoClient mongoClient =
        MongoClient.create("mongodb://localhost:27017");

        // Access database

        MongoDBDatabase database = mongoClient.getDatabase("CollegeDB");

        // Access collection

        MongoCollection<Document> collection =
        database.getCollection("Students");

        // =====
        // CREATE (Insert)
        // =====

        Document student = new Document("id", 103)
        .append("name", "Joy")
        .append("department", "Electrical")
        .append("cgpa", 7.9);
```

```

collection.insertOne(student);

System.out.println("Document Inserted Successfully");

// =====

// READ (Find)

// =====

System.out.println("\nStudent Records:");

FindIterable<Document> students = collection.find();

for (Document doc : students) {

System.out.println(doc.toJson());

}

// // =====

// // UPDATE

// // =====

collection.updateOne(

Filters.eq("id", 101),

Updates.set("cgpa", 9.2)

);

System.out.println("\nDocument Updated Successfully");

// // =====

// // READ Updated Record

// // =====

Document updatedStudent =

collection.find(Filters.eq("id", 101)).first();

System.out.println("\nUpdated Record:");

System.out.println(updatedStudent.toJson());

// // =====

// // DELETE

// // =====

```

```
collection.deleteOne(Filters.eq("id", 101));

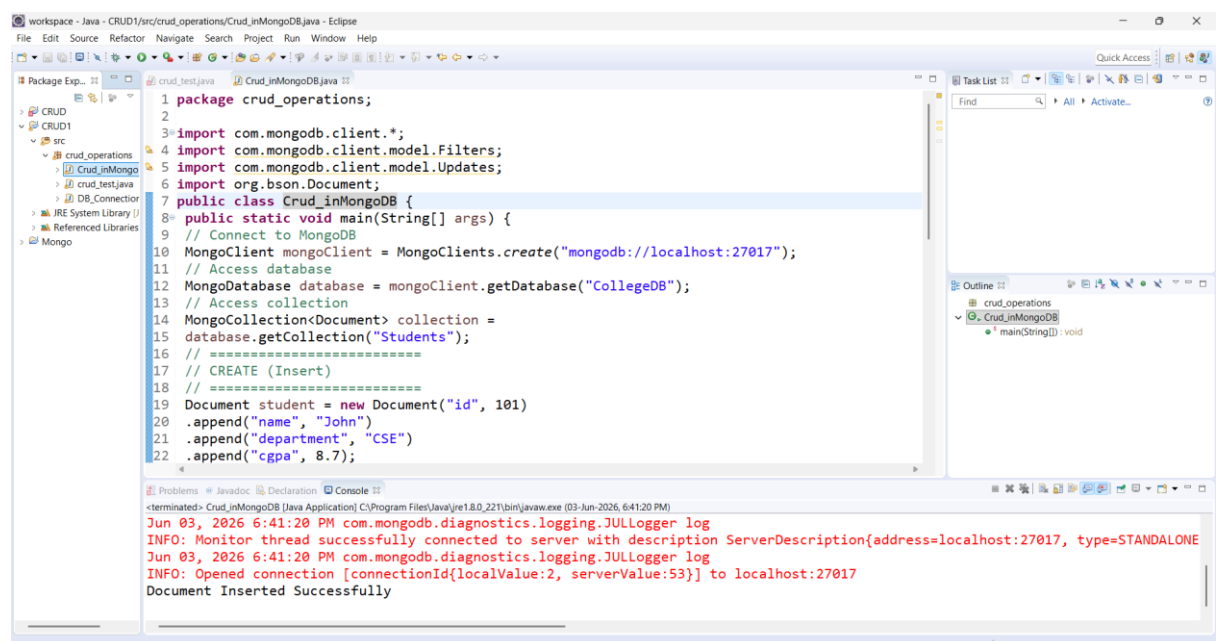
System.out.println("\nDocument Deleted Successfully");

// Close connection
mongoClient.close();
}
}
```

2. Initial Database and Collection

```
>_MONGOSH

> use CollegeDB
< switched to db CollegeDB
> db.Students.insertOne({
  id: 100,
  name: "Jacob",
  department: "CSE",
  cgpa: 8.8
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a20266945be2643af04cdad')
}
> db.Students.find()
< {
  _id: ObjectId('6a20266945be2643af04cdad'),
  id: 100,
  name: 'Jacob',
  department: 'CSE',
  cgpa: 8.8
}
> db.Students.find()
< {
  _id: ObjectId('6a20266945be2643af04cdad'),
  id: 100,
  name: 'Jacob',
  department: 'CSE',
  cgpa: 8.8
}
{
```



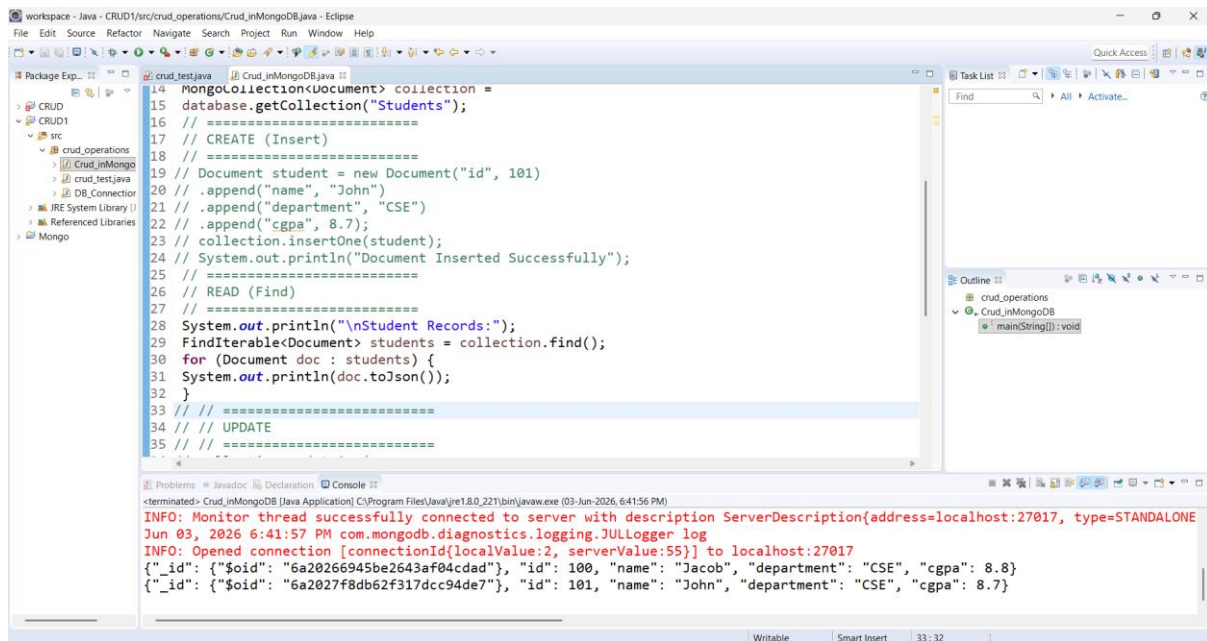
3.Create():

```

> db.Students.find()
< {
  _id: ObjectId('6a20266945be2643af04cdad'),
  id: 100,
  name: 'Jacob',
  department: 'CSE',
  cgpa: 8.8
}
{
  _id: ObjectId('6a2027f8db62f317dcc94de7'),
  id: 101,
  name: 'John',
  department: 'CSE',
  cgpa: 8.7
}
CollegeDB>

```

4.Read():



```
workspace - Java - CRUD1/src/crud_operations/Crud_inMongoDB.java - Eclipse
File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer:
  > crud_operations
    > crud_inMongoDB
      > crud_test.java
      > DB_Connector
  > JRE System Library
  > Referenced Libraries
  > Mongo

Main Editor:
14 MongoClient<Document> collection =
15 database.getCollection("Students");
16 // =====
17 // CREATE (Insert)
18 // =====
19 // Document student = new Document("id", 101)
20 // .append("name", "John")
21 // .append("department", "CSE")
22 // .append("cgpa", 8.7);
23 // collection.insertOne(student);
24 // System.out.println("Document Inserted Successfully");
25 // =====
26 // READ (Find)
27 // =====
28 System.out.println("\nStudent Records:");
29 FindIterable<Document> students = collection.find();
30 for (Document doc : students) {
31 System.out.println(doc.toJson());
32 }
33 // =====
34 // UPDATE
35 // =====

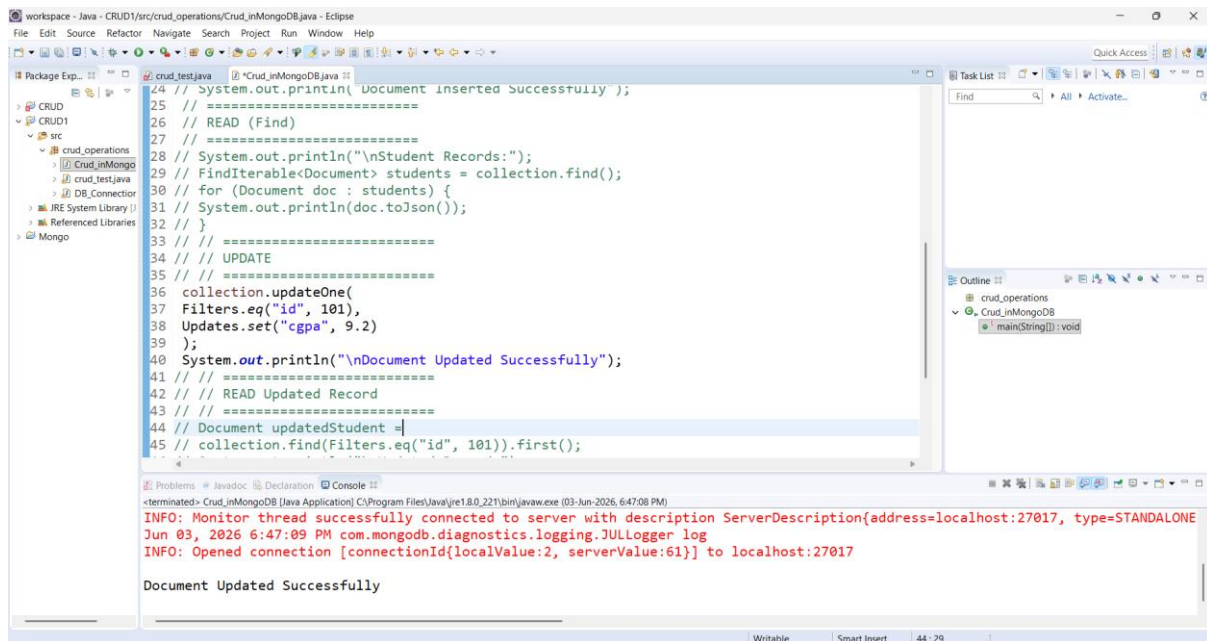
Task List:
Find
All
Activate...

Outline:
  > crud_operations
    > Crud_inMongoDB
      > main(String[]) : void

Console:
<terminated> Crud_inMongoDB [Java Application] C:\Program Files\Java\jre1.8.0_221\bin\javaw.exe (03-Jun-2026, 6:41:56 PM)
INFO: Monitor thread successfully connected to server with description ServerDescription{address=localhost:27017, type=STANDALONE
Jun 03, 2026 6:41:57 PM com.mongodb.diagnostics.logging.JULLogger log
INFO: Opened connection [connectionId{localValue:2, serverValue:55}] to localhost:27017
{"_id": {"$oid": "6a20266945be2643af04cdad"}, "id": 100, "name": "Jacob", "department": "CSE", "cgpa": 8.8}
{"_id": {"$oid": "6a2027f8db62f317dcc94de7"}, "id": 101, "name": "John", "department": "CSE", "cgpa": 8.7}
```

```
> db.Students.find()
< {
  _id: ObjectId('6a20266945be2643af04cdad'),
  id: 100,
  name: 'Jacob',
  department: 'CSE',
  cgpa: 8.8
}
{
  _id: ObjectId('6a2027f8db62f317dcc94de7'),
  id: 101,
  name: 'John',
  department: 'CSE',
  cgpa: 8.7
}
CollegeDB >
```

5.Update():



The screenshot shows the Eclipse IDE with a Java project named 'CRUD1'. The package explorer on the left shows the project structure: 'CRUD1' contains 'src', which contains 'crud_operations' and 'crud_test.java'. The 'crud_operations' package contains 'Crud_inMongoDB.java'. The 'main' method in 'Crud_inMongoDB.java' is shown, which performs an update operation. The code is as follows:

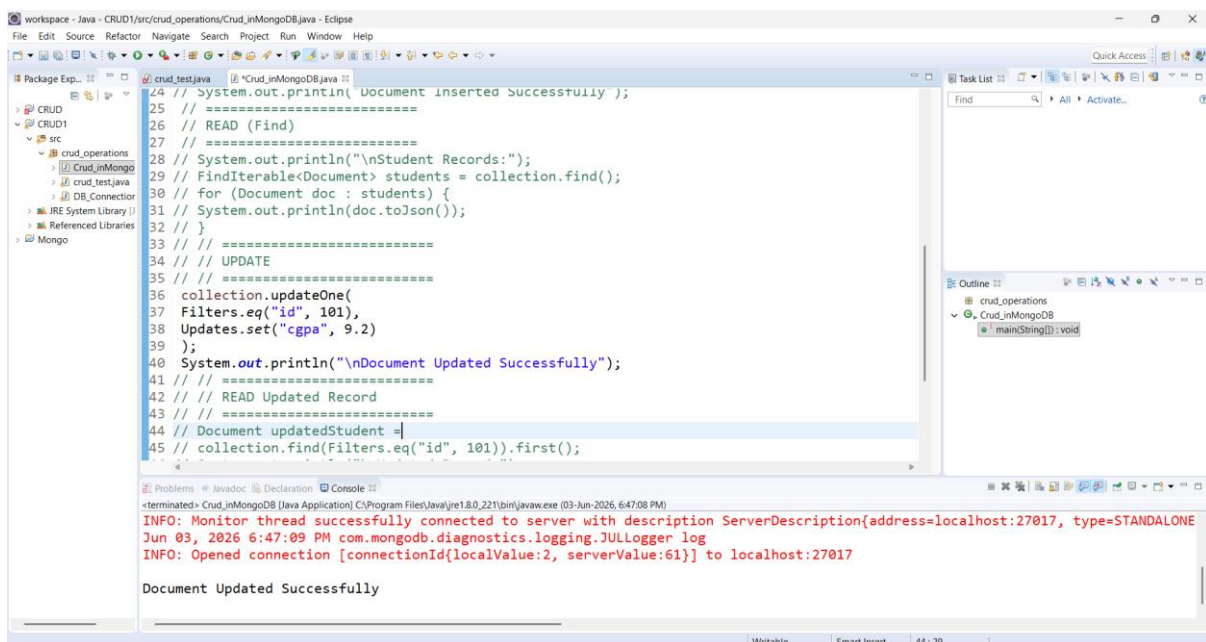
```
24 // system.out.println( Document inserted Successfully );
25 // =====
26 // READ (Find)
27 // =====
28 // System.out.println("\nStudent Records:");
29 // FindIterable<Document> students = collection.find();
30 // for (Document doc : students) {
31 // System.out.println(doc.toJson());
32 // }
33 // // =====
34 // // UPDATE
35 // // =====
36 collection.updateOne(
37     Filters.eq("id", 101),
38     Updates.set("cgpa", 9.2)
39 );
40 System.out.println("\nDocument Updated Successfully");
41 // // =====
42 // // READ Updated Record
43 // // =====
44 // Document updatedStudent =
45 // collection.find(Filters.eq("id", 101)).first();
```

The console output shows the following messages:

```
<terminated> Crud_inMongoDB [Java Application] C:\Program Files\Java\jre1.8.0_221\bin\javaw.exe (03-Jun-2026, 6:47:08 PM)
INFO: Monitor thread successfully connected to server with description ServerDescription{address=localhost:27017, type=STANDALONE
Jun 03, 2026 6:47:09 PM com.mongodb.diagnostics.logging.JULLogger log
INFO: Opened connection [connectionId{localValue:2, serverValue:61}] to localhost:27017

Document Updated Successfully
```

6.ReadUpdated():



The screenshot shows the Eclipse IDE with the same Java project as in the previous image. The package explorer on the left shows the project structure: 'CRUD1' contains 'src', which contains 'crud_operations' and 'crud_test.java'. The 'crud_operations' package contains 'Crud_inMongoDB.java'. The 'main' method in 'Crud_inMongoDB.java' is shown, which performs a read updated operation. The code is as follows:

```
24 // system.out.println( Document inserted Successfully );
25 // =====
26 // READ (Find)
27 // =====
28 // System.out.println("\nStudent Records:");
29 // FindIterable<Document> students = collection.find();
30 // for (Document doc : students) {
31 // System.out.println(doc.toJson());
32 // }
33 // // =====
34 // // UPDATE
35 // // =====
36 collection.updateOne(
37     Filters.eq("id", 101),
38     Updates.set("cgpa", 9.2)
39 );
40 System.out.println("\nDocument Updated Successfully");
41 // // =====
42 // // READ Updated Record
43 // // =====
44 // Document updatedStudent =
45 // collection.find(Filters.eq("id", 101)).first();
```

The console output shows the following messages:

```
<terminated> Crud_inMongoDB [Java Application] C:\Program Files\Java\jre1.8.0_221\bin\javaw.exe (03-Jun-2026, 6:47:08 PM)
INFO: Monitor thread successfully connected to server with description ServerDescription{address=localhost:27017, type=STANDALONE
Jun 03, 2026 6:47:09 PM com.mongodb.diagnostics.logging.JULLogger log
INFO: Opened connection [connectionId{localValue:2, serverValue:61}] to localhost:27017

Document Updated Successfully
```

```

> db.Students.find()
< {
  _id: ObjectId('6a20266945be2643af04cdad'),
  id: 100,
  name: 'Jacob',
  department: 'CSE',
  cgpa: 8.8
}
{
  _id: ObjectId('6a2027f8db62f317dcc94de7'),
  id: 101,
  name: 'John',
  department: 'CSE',
  cgpa: 9.2
}

```

7.Delete():

The screenshot shows the Eclipse IDE with a Java project named 'crud_operations'. The main class is 'Crud_inMongoDB.java'. The code performs the following steps:

- Connects to MongoDB at localhost:27017.
- Updates a document with id 101, setting cgpa to 9.2.
- Prints the updated document to the console.
- Deletes the document with id 101.
- Prints 'Document Deleted Successfully' to the console.
- Closes the MongoDB connection.

The console output shows the following messages:

```

INFO: Monitor thread successfully connected to server with description ServerDescription(address=localhost:27017, type=STANDALONE
Jun 03, 2026 6:51:42 PM com.mongodb.diagnostics.logging.JULLogger log
INFO: Opened connection [connectionId{localValue:2, serverValue:65}] to localhost:27017

Document Deleted Successfully

```

8. Insert Many Documents

```
17 // CREATE (Insert)
18 // =====
19 Document student1 = new Document("id", 100)
20     .append("name", "Jacob")
21     .append("department", "CSE")
22     .append("cgpa", 8.8);
23
24 Document student2 = new Document("id", 102)
25     .append("name", "Jean")
26     .append("department", "Mech")
27     .append("cgpa", 8.2);
28
29 Document student3 = new Document("id", 103)
30     .append("name", "Joy")
31     .append("department", "Electrical")
32     .append("cgpa", 7.9);
33
34 collection.insertMany(java.util.Arrays.asList(
35     student1,
36     student2,
37     student3
38 ));
39
40 System.out.println("Documents Inserted Successfully");
41
42 // =====
```

```
> db.Students.find()
< {
  _id: ObjectId('6a20266945be2643af04cdad'),
  id: 100,
  name: 'Jacob',
  department: 'CSE',
  cgpa: 8.8
}
{
  _id: ObjectId('6a2027f8db62f317dcc94de7'),
  id: 101,
  name: 'John',
  department: 'CSE',
  cgpa: 8.7
}
{
  _id: ObjectId('6a2028a776efde7295e95854'),
  id: 102,
  name: 'Jean',
  department: 'Mech',
  cgpa: 8.2
}
{
  _id: ObjectId('6a2028e01c47697a58f45cf4'),
  id: 103,
  name: 'Joy',
  department: 'Electrical',
  cgpa: 7.9
}
```

9. Find Specific Document:

```
45 Document doc = collection.find(Filters.eq("id", 102)).first();
46 System.out.println(doc.toJson());
```

```
INFO: Opened connection [connectionId{localValue:1, serverValue:68}] to localhost:27017
Jun 03, 2026 7:52:49 PM com.mongodb.diagnostics.logging.JULLogger log
INFO: Monitor thread successfully connected to server with description ServerDescription{address=localhost:27017, type=STANDALONE}
Jun 03, 2026 7:52:49 PM com.mongodb.diagnostics.logging.JULLogger log
INFO: Opened connection [connectionId{localValue:2, serverValue:69}] to localhost:27017
{"_id": {"$oid": "6a2028a776efde7295e95854"}, "id": 102, "name": "Jean", "department": "Mech", "cgpa": 8.2}
```

10. Update Specific Document

```
58 collection.updateOne(
59     Filters.eq("id", 102),
60     Updates.set("name", "Bobby")
61 );
62
63 System.out.println("Document Updated Successfully");
```

```
<terminated> Crud_inMongoDB [Java Application] C:\Program Files\Java\jre1.8.0_221\bin\javaw.exe (03-Jun-2026, 7:57:23 PM)
INFO: Opened connection [connectionId{localValue:1, serverValue:70}] to localhost:27017
Jun 03, 2026 7:57:23 PM com.mongodb.diagnostics.logging.JULLogger log
INFO: Monitor thread successfully connected to server with description ServerDescription{address=localhost:27017, type=STANDALONE}
Jun 03, 2026 7:57:23 PM com.mongodb.diagnostics.logging.JULLogger log
INFO: Opened connection [connectionId{localValue:2, serverValue:71}] to localhost:27017
Document Updated Successfully
```

11. Delete Specific Document:

```
51 collection.deleteOne(Filters.eq("id", 101));
52 System.out.println("\nDocument Deleted Successfully");
53 // Close connection
54 MongoClient.close();
55 }
56 }
```

```
Problems Javadoc Declaration Console
<terminated> Crud_inMongoDB [Java Application] C:\Program Files\Java\jre1.8.0_221\bin\javaw.exe (03-Jun-2026, 6:51:41 PM)
INFO: Monitor thread successfully connected to server with description ServerDescription{address=localhost:27017, type=STANDALONE}
Jun 03, 2026 6:51:42 PM com.mongodb.diagnostics.logging.JULLogger log
INFO: Opened connection [connectionId{localValue:2, serverValue:65}] to localhost:27017

Document Deleted Successfully
```